

Ali Sufiyan Khan

VIT Vellore | alizaydsab@gmail.com | linkedin.com/in/ali-khan-4197b1217 | github.com/sting-raider

Education

Vellore Institute of Technology (VIT Vellore), B.Tech in Computer Science and Engineering Aug 2023 – May 2027 (Expected)

- CGPA: 7.96/10.0
- **Relevant Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Artificial Intelligence

Projects

F1 Reinforcement Learning Agent github.com/sting-raider/f1-rl-agent

- Engineered a high-performance autonomous racing agent for F1 22 using Python and Stable Baselines3 (PPO), utilizing an MLP policy on raw telemetry data for optimal training efficiency.
- Developed a low-latency UDP packet parser using Python socket and struct to decode game physics (60Hz) directly, eliminating external middleware dependencies.
- Designed a custom Gymnasium environment with a "Virtual Lidar" system, utilizing vector mathematics to calculate lateral track deviation and lookahead trajectory from recorded track maps.
- Implemented continuous action space control via a virtual Xbox 360 controller driver (vgamepad), enabling precise analog steering and throttle modulation compared to discrete keyboard inputs.
- Optimized the training pipeline with busy-wait synchronization to support 3x-5x game simulation speeds, significantly reducing model convergence time.

Context-Aware RAG Agent github.com/sting-raider/AI-ML-PROJECT

- Built and trained a Convolutional Neural Network (CNN) for image classification on the CIFAR-10 dataset, implementing data augmentation and hyperparameter tuning.
- Developed an AI assistant using the LangChain framework, integrating Retrieval-Augmented Generation (RAG) to answer questions from a local document knowledge base.
- Implemented function calling within the LangChain agent to perform external tasks such as arithmetic calculations.

Technologies

Languages: Python, C/C++, Java, Rust (learning currently)

Frameworks & Libraries: PyTorch, TensorFlow, LangChain, Scikit-learn, OpenAI Gym, Stable Baselines3, OpenCV, PyAutoGUI

Developer Tools: Git & GitHub, Docker, MySQL, JetBrains IDEs (CLion, IntelliJ, PyCharm, etc.)